WORKED SOLUTION

STEP III 2000, Question 3

Pure • Complex Numbers and Geometry

STEP III 2000, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s3-q3

This question has a full guided treatment in *Beyond STEP*, Section 5.6, *The Fermat Point*:
https://beyondstep.keithmcnulty.org/ch05/fermat-point.html

The key idea

In the Argand plane, rotating a vector through 60° is just multiplication by $\alpha = e^{i\pi/3}$, so the third vertex of an equilateral triangle on a given side can be written down in one line. After that, everything comes from the identity $1 + \alpha^2 = \alpha$ and its relatives, such as $\alpha^3 = -1$. They collapse the vertex expressions into very simple forms, and once you see those, the final part is almost immediate.

Solution

The identity

Since $\alpha^2 = e^{2i\pi/3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$,

$$1 + \alpha^2 = \tfrac{1}{2} + \tfrac{\sqrt{3}}{2}i = \cos\tfrac{\pi}{3} + i\sin\tfrac{\pi}{3} = \alpha.$$

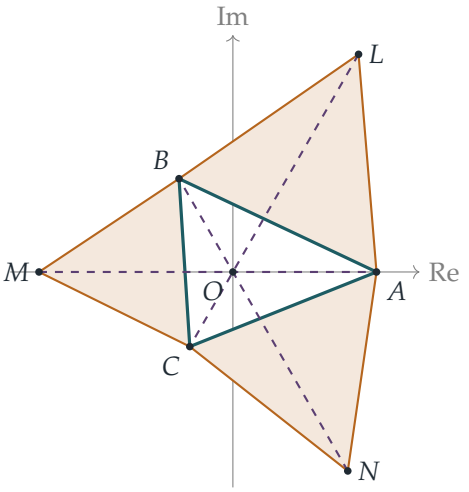
We shall also use these consequences:

- $\alpha^3 = e^{i\pi} = -1$;
- $1 - \alpha = -\alpha^2$ (this is the identity rearranged);
- $\alpha^2 - \alpha = -1$ (also a rearrangement);
- $1 - \alpha = \bar{\alpha} = e^{-i\pi/3}$, since $\alpha + \bar{\alpha} = 2\cos\frac{\pi}{3} = 1$.

The last one says that multiplying by $1 - \alpha$ rotates through 60° *clockwise*.

The triangle ABC

$A = p$ lies on the positive real axis, $B = q\alpha^2 = q\,\mathrm{e}^{2\mathrm{i}\pi/3}$ lies on the ray at argument 120°, and $C = -r\alpha = r\,\mathrm{e}^{4\mathrm{i}\pi/3}$ lies on the ray at argument 240°. So the three vertices sit on three rays from O that are 120° apart, at distances p, q, r from O. The origin is inside the triangle, and $A \to B \to C$ goes *anticlockwise*. The sketch below uses $p = 2$, $q = 1.5$, $r = 1.2$, and shows the erected triangles too.



The vertices N, L and M

If XYZ is an equilateral triangle lettered clockwise, then going round from X, the side XZ is the side XY turned clockwise through 60° about X. So

$$Z - X = (Y - X)\,\mathrm{e}^{-\mathrm{i}\pi/3} = (Y - X)(1 - \alpha), \qquad \text{that is,} \qquad Z = \alpha X + (1 - \alpha)Y.$$

Ask yourself Why does "clockwise" put the new triangles *outside* ABC?

Since $A \to B \to C$ is anticlockwise, C lies to the left of the directed side AB. Turning AB clockwise about A swings it to the right, so L lies on the opposite side of AB from C. The same holds for the other two sides, which matches the picture.

N (triangle CAN, so $X = C$, $Y = A$):

$$N = \alpha(-r\alpha) + (1 - \alpha)p = (1 - \alpha)p - \alpha^2 r,$$

as required. Using $1 - \alpha = -\alpha^2$, this tidies to $N = -\alpha^2(p + r)$.

L (triangle ABL, so $X = A$, $Y = B$):

$$L = \alpha p + (1 - \alpha)q\alpha^2 = \alpha p + q(\alpha^2 - \alpha^3) = \alpha p + q(1 + \alpha^2) = (p + q)\alpha,$$

using $\alpha^3 = -1$ and then $1 + \alpha^2 = \alpha$.

M (triangle BCM, so $X = B$, $Y = C$):

$$M = \alpha \cdot q\alpha^2 + (1-\alpha)(-r\alpha) = q\alpha^3 + r(\alpha^2 - \alpha) = -q - r = -(q+r),$$

using $\alpha^3 = -1$ and $\alpha^2 - \alpha = -1$.

So, in the same style as the given form of N,

$$L = \alpha p + \alpha q, \qquad M = -q - r, \qquad N = (1-\alpha)p - \alpha^2 r.$$

The three lines

Now look at each pair of points as a positive multiple of a unit complex number:

$$\begin{aligned} L &= (p+q)\,\mathrm{e}^{\mathrm{i}\pi/3}, & C &= -r\alpha = r\,\mathrm{e}^{\mathrm{i}\pi/3}\cdot(-1), \\ M &= (q+r)\cdot(-1), & A &= p\cdot 1, \\ N &= (p+r)\cdot(-\alpha^2), & B &= q\,\alpha^2. \end{aligned}$$

In each pair the two points are positive real multiples of u and $-u$ for a unit complex number u (namely $u = \alpha$, $u = 1$ and $u = \alpha^2$). So the two points lie on a common line through O, on *opposite* sides of O. Hence each of the lines LC, MA, NB passes through the origin, and in fact the origin lies on each segment. Then, because the points are on opposite sides of O, the length of each segment is the sum of the distances from O:

$$LC = (p+q) + r, \qquad MA = (q+r) + p, \qquad NB = (p+r) + q,$$

all equal to $p + q + r$, as required.

Checks

With $p = 1$, $q = 2$, $r = 3$ the formulas give $L = 3\alpha \approx 1.5 + 2.598\mathrm{i}$, $M = -5$ and $N = -4\alpha^2 \approx 2 - 3.464\mathrm{i}$. Measuring $|L - A| = |L - B| = |A - B| = \sqrt{7}$ confirms that ABL is equilateral. A computer built each triangle directly from the two rotations by $\pm 60^\circ$, kept the clockwise one, and confirmed the three formulas, the concurrence at O and the common length $p + q + r$ for 10 000 random choices of p, q, r. A nice special case: when $p = q = r$, triangle ABC is equilateral with centre O, and the figure has the expected threefold symmetry.

Common slips

- Rotating the wrong way, by α instead of $\bar{\alpha} = 1 - \alpha$. That puts L, M, N inside the triangle, and the given form of N won't come out.
- Showing only that O is on each line and forgetting that the length then needs O to lie *between* the endpoints. The signs of the real multiples settle this.

Beyond STEP

This is a special case of Napoleon's configuration: for *any* triangle with equilateral triangles erected outwards, the lines AM, BN, CL meet at a single point (the Fermat point) and have equal lengths. The Fermat point minimises the total distance to the three vertices; here it is O, where each side subtends $120°$. Complex numbers turn rotations into multiplications, and that is why proofs like this become short algebra. Needham's *Visual Complex Analysis* develops this geometric viewpoint beautifully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.